

```
C:\WINDOWS\system32\cmd. x Windows PowerShell x Windows PowerShell x Windows PowerShell x + v - o x

(venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking>notepad mlflow_experiment.py

(venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking>python mlflow_experiment.py
2026/09/16 17:47:24 INFO mlflow.store.db.utils: Creating initial MLflow database tables...
2026/09/16 17:47:24 INFO mlflow.store.db.utils: Updating database tables
2026/09/16 17:47:28 INFO mlflow.tracking.fluent: Experiment with name 'W11D1_MLflow_Experiment_Tracking' does not exist. Creating a new experiment.
=====
W11D1 - MLFLOW EXPERIMENT TRACKING
=====

Running Experiment 1
-----
Parameters: {'n_estimators': 50, 'max_depth': 3, 'min_samples_split': 2}
Accuracy : 0.9667
Precision: 0.9697
Recall : 0.9667
F1 Score : 0.9666
Run ID : 8ef606b5901d4d01a082056d14279284

Running Experiment 2
-----
Parameters: {'n_estimators': 100, 'max_depth': 5, 'min_samples_split': 2}
Accuracy : 0.9333
Precision: 0.9333
Recall : 0.9333
F1 Score : 0.9333
Run ID : 4373eca52c89478ba886a3086b9292a0

Running Experiment 3
-----
Parameters: {'n_estimators': 150, 'max_depth': 7, 'min_samples_split': 2}
Accuracy : 0.9000
Precision: 0.9024
Recall : 0.9000
F1 Score : 0.8997
```

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Run ID : 7002df35a4df4e28895c757282f7b6ab

Running Experiment 4
-----
Parameters: {'n_estimators': 200, 'max_depth': 10, 'min_samples_split': 2}
Accuracy : 0.9000
Precision: 0.9024
Recall : 0.9000
F1 Score : 0.8997
Run ID : 0ee081126d2d479dba1557679f1af114

Running Experiment 5
-----
Parameters: {'n_estimators': 100, 'max_depth': None, 'min_samples_split': 4}
Accuracy : 0.9667
Precision: 0.9697
Recall : 0.9667
F1 Score : 0.9666
Run ID : 521f0e04667a4c28803707d449897f2b

=====
BEST EXPERIMENT
=====
Best Accuracy: 0.9667
Best Run ID : 8ef606b5901d4d01a082056d14279284
Best Params : {'n_estimators': 50, 'max_depth': 3, 'min_samples_split': 2}

Registering best model...
Model URI: runs:/8ef606b5901d4d01a082056d14279284/model
Successfully registered model 'W11D1_RandomForest_Best_Model'.
2026/09/16 17:50:00 WARNING mlflow.tracking._model_registry.fluent: Run with id 8ef606b5901d4d01a082056d14279284 has no artifacts at artifact path 'model', registering model based on models:/m-f8383e544d49418d959ec758997e44f2 instead
Created version '1' of model 'W11D1_RandomForest_Best_Model'.

=====
MODEL REGISTRATION COMPLETE
=====
```

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Recall : 0.9000
F1 Score : 0.8997
Run ID : 0ee081126d2d479dba1557679f1af114

Running Experiment 5
=====
Parameters: {'n_estimators': 100, 'max_depth': None, 'min_samples_split': 4}
Accuracy : 0.9667
Precision: 0.9697
Recall : 0.9667
F1 Score : 0.9666
Run ID : 521f0e04667a4c28803707d449897f2b

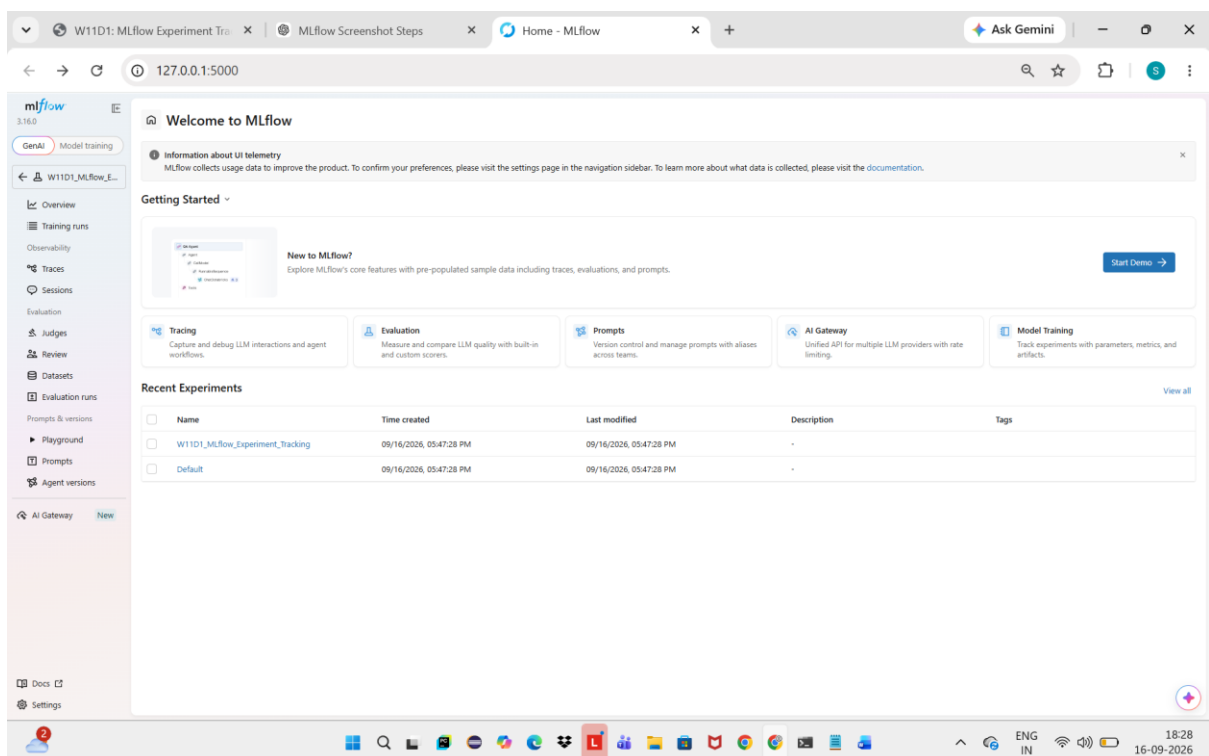
=====
BEST EXPERIMENT
=====
Best Accuracy: 0.9667
Best Run ID : 8ef606b5901d4d01a082056d14279284
Best Params : {'n_estimators': 50, 'max_depth': 3, 'min_samples_split': 2}

Registering best model...
Model URI: runs:/8ef606b5901d4d01a082056d14279284/model
Successfully registered model 'W11D1_RandomForest_Best_Model'.
2026/09/16 17:50:00 WARNING mlflow.tracking._model_registry.fluent: Run with id 8ef606b5901d4d01a082056d14279284 has no artifacts at
artifact path 'model', registering model based on models:/m-f8383e544d49418d959ec758997e44f2 instead
Created version '1' of model 'W11D1_RandomForest_Best_Model'.

=====
MODEL REGISTRATION COMPLETE
=====
Model Name : W11D1_RandomForest_Best_Model
Model Version: 1
Best Run ID : 8ef606b5901d4d01a082056d14279284

MLflow experiment completed successfully.

(env) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking>
```



W11D1: MLflow Experiment Tracking | MLflow Screenshot Steps | Runs - Experiment 1 - MLflow | Ask Gemini

127.0.0.1:5000/#/experiments/1/runs?workflowType=genai&searchFilter=&orderByKey=attributes.start_time&orderByAsc=false&startTime=ALL...

Training runs

Search: metrics.rmse < 1 and params.model = "tree" | All time | State: Active | Datasets | Sort: Created | Columns | Group by | + New run

Run Name	Created	Dataset	Duration	Source	Models	Metrics
experiment_5	39 minutes ago	-	25.9s	mlflow_e...	model	accuracy: 0.96666666...
experiment_4	40 minutes ago	-	28.1s	mlflow_e...	model	0.9
experiment_3	40 minutes ago	-	26.2s	mlflow_e...	model	0.9
experiment_2	41 minutes ago	-	25.1s	mlflow_e...	model	0.93333333...
experiment_1	41 minutes ago	-	44.3s	mlflow_e...	W11D1_RandomForest_Be...	0.96666666...

5 matching runs

W11D1: MLflow Experiment Tracking | Overview | Training runs | Observability | Traces | Sessions | Evaluation | Judges | Review | Datasets | Evaluation runs | Prompts & versions | Playground | Prompts | Agent versions | AI Gateway | New

W11D1: MLflow Experiment Tracking | MLflow Screenshot Steps | Run 8ef606b5901d4d01a082056d14279284 | Ask Gemini

127.0.0.1:5000/#/experiments/1/runs/8ef606b5901d4d01a082056d14279284

experiment_1

Overview | Model metrics | System metrics | Traces | Artifacts

Description: No description

Metrics (4)

Metric	Value	Model
accuracy	0.9666666666666667	model
precision	0.9666666666666667	model
recall	0.9666666666666667	model
f1_score	0.9666666666666667	model

Parameters (3)

Parameter	Value
n_estimators	50
max_depth	3
min_samples_split	2

Logged models (1)

Type	Step	Model name	Status	Created	Registered models	Dataset	accuracy	precision	recall	f1_score
Output	0	model	Ready	42 minutes ago	W11D1_RandomForest_Best	-	0.9666666666666667	0.9666666666666667	0.9666666666666667	0.9666666666666667

About this run

Created at: 09/16/2026, 09:47:29 PM
Created by: Saranya CA
Experiment ID: 1
Status: Finished
Run ID: 8ef606b5901d4d01a082056d14279284
Duration: 44.3s
Source: mlflow_experiment.py
Registered prompts: --
Datasets: None
Tags: dataset: H11 framework: scikit-learn experiment_number: 1
Registered models: W11D1_RandomForest_Best_Model v1

W11D1: MLflow Experiment Tr... MLflow Screenshot Steps Run 4373eca52c89478ba886a3086b9292a0

127.0.0.1:5000/#/experiments/1/runs/4373eca52c89478ba886a3086b9292a0

mlflow 3.10.0

Model training

Overview Model metrics System metrics Traces Artifacts

Description No description

Metrics (4)

Metric	Value	Model
accuracy	0.9333333333333333	model
precision	0.9333333333333333	model
recall	0.9333333333333333	model
f1_score	0.9333333333333333	model

Parameters (3)

Parameter	Value
n_estimators	100
max_depth	5
min_samples_split	2

Logged models (1)

Type	Step	Model name	Status	Created	Registered models	Dataset	accuracy	precision	recall	f1_score
Output	0	model	Ready	41 minutes ago	-	-	0.9333333333333333	0.9333333333333333	0.9333333333333333	0.9333333333333333

About this run

Created at 09/16/2026 09:48:14 PM

Created by Sergana CA

Experiment ID 1

Status Finished

Run ID 4373eca52c89478ba886a3086b9292a0

Duration 25.1s

Source mlflow_experiment.py

Registered prompts -

Dataset None

Tags dataset: iris framework: scikit-learn experiment_number: 1

Registered models None

W11D1: MLflow Experiment Tr... MLflow Screenshot Steps Run 7002df35a4df4e28895c757282f7b6ab

127.0.0.1:5000/#/experiments/1/runs/7002df35a4df4e28895c757282f7b6ab

mlflow 3.10.0

Model training

Overview Model metrics System metrics Traces Artifacts

Description No description

Metrics (4)

Metric	Value	Model
accuracy	0.9	model
precision	0.9023989023989024	model
recall	0.9	model
f1_score	0.8997487743333333	model

Parameters (3)

Parameter	Value
n_estimators	150
max_depth	7
min_samples_split	2

Logged models (1)

Type	Step	Model name	Status	Created	Registered models	Dataset	accuracy	precision	recall	f1_score
Output	0	model	Ready	41 minutes ago	-	-	0.9	0.9023989023989024	0.9	0.8997487743333333

About this run

Created at 09/16/2026 09:48:39 PM

Created by Sergana CA

Experiment ID 1

Status Finished

Run ID 7002df35a4df4e28895c757282f7b6ab

Duration 26.2s

Source mlflow_experiment.py

Registered prompts -

Dataset None

Tags dataset: iris framework: scikit-learn experiment_number: 1

Registered models None

W11D1: MLflow Experiment Tr... MLflow Screenshot Steps Run 0ee081126d2d479dba1557af1af114

127.0.0.1:5000/#/experiments/1/runs/0ee081126d2d479dba1557af1af114

mlflow 3.10.0

experiment_4

Overview Model metrics System metrics Traces Artifacts

Description

Metrics (4)

Metric	Value	Model
accuracy	0.9	model
precision	0.9023986023986024	model
recall	0.9	model
f1_score	0.8947460734335839	model

Parameters (3)

Parameter	Value
n_estimators	200
max_depth	10
min_samples_split	2

Logged models (1)

Type	Step	Model name	Status	Created	Registered models	Dataset	accuracy	precision	recall	f1_score
Output	0	model	Ready	41 minutes ago	-	-	0.9	0.9023986023986024	0.9	0.8947460734335839

About this run

Created at: 09/16/2026, 09:48:05 PM
Created by: Sergana CA
Experiment ID: 1
Status: Finished
Run ID: 0ee081126d2d479dba1557af1af114
Duration: 28.1s
Source: mlflow_experiment.py
Registered prompts: --

Tags

dataset: iris framework: scikit-learn experiment_number: 4

Registered models

None

W11D1: MLflow Experiment Tr... MLflow Screenshot Steps Run 521f0e04667a4c28803707d449897f2b

127.0.0.1:5000/#/experiments/1/runs/521f0e04667a4c28803707d449897f2b

mlflow 3.10.0

experiment_5

Overview Model metrics System metrics Traces Artifacts

Description

Metrics (4)

Metric	Value	Model
accuracy	0.9505531244778813	model
precision	0.9505531244778813	model
recall	0.9505531244778813	model
f1_score	0.9505531244778813	model

Parameters (3)

Parameter	Value
n_estimators	100
max_depth	None
min_samples_split	4

Logged models (1)

Type	Step	Model name	Status	Created	Registered models	Dataset	accuracy	precision	recall	f1_score
Output	0	model	Ready	41 minutes ago	-	-	0.9505531244778813	0.9505531244778813	0.9505531244778813	0.9505531244778813

About this run

Created at: 09/16/2026, 09:48:34 PM
Created by: Sergana CA
Experiment ID: 1
Status: Finished
Run ID: 521f0e04667a4c28803707d449897f2b
Duration: 25.9s
Source: mlflow_experiment.py
Registered prompts: --

Tags

dataset: iris framework: scikit-learn experiment_number: 5

Registered models

None

```
C:\WINDOWS\system32\cmd.exe X Windows PowerShell X Windows PowerShell X Windows PowerShell X + -
mlmodel_local_path = _download_artifact_from_uri(artifact_uri=mlmodel_file_path)
#####
File "C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking\venv\Lib\site-packages\mlflow\tracking\artifact_utils.py", line 128, in _download_artifact_from_uri
    return repo.download_artifacts(artifact_path=artifact_path, dst_path=output_path)
#####
File "C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking\venv\Lib\site-packages\mlflow\store\artifact\local_artifact_repo.py", line 247, in download_artifacts
    return self.get_local_path(artifact_path)
#####
File "C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking\venv\Lib\site-packages\mlflow\store\artifact\local_artifact_repo.py", line 70, in get_local_path
    raise MLflowException(
mlflow.exceptions.MLflowException: No such artifact: 'MLmodel'
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking> ^C
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking> echo $env:MLFLOW_TRACKING_URI
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking> Remove-Item Env:MLFLOW_TRACKING_URI -ErrorAction SilentlyContinue
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking> $env:MLFLOW_TRACKING_URI="sqlite:///mlflow.db"
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking> echo $env:MLFLOW_TRACKING_URI
sqlite:///mlflow.db
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking> mlflow artifacts download -u "runs:/Bef606b5901d4d01a082056d14279284/model" -d ".\served_model"

C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking\served_model\model
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking> mlflow models serve -m ".\served_model\model" -p 5001 --no-conda
2026/09/16 18:05:27 INFO mlflow.models.flavor_backend_registry: Selected backend for flavor 'python_function'
2026/09/16 18:05:27 INFO mlflow.pyfunc.backend: === Running command 'uvicorn --host 127.0.0.1 --port 5001 --workers 1 mlflow.pyfunc.serving_server.app:app'
INFO: Started server process [4024]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://127.0.0.1:5001 (Press CTRL+C to quit)
INFO: 127.0.0.1:64845 - "POST /invocations HTTP/1.1" 200 OK
```

```
C:\WINDOWS\system32\cmd.exe X Windows PowerShell X Windows PowerShell X Windows PowerShell X + -
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Sanjana CA> cd "C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking"
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking> venv\Scripts\activate
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking> python -c "import mlflow; m=mlflow.pyfunc.load_model(r'.\served_model\model'); print(m.metadata.get_input_schema())"
[Tensor('float64', (-1, 4))]
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking> $body = '{"inputs":[[5.1,3.5,1.4,0.2]]}'
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking> Invoke-RestMethod -Uri "http://127.0.0.1:5001/invocations" -Method Post -ContentType "application/json" -Body $body

predictions
-----
{0}

(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D1_MLflow_Experiment_Tracking>
```